

## Response Letter with Annotations

Toubia, O., Berger, J., & Eliashberg, J. (2021). How quantifying the shape of stories predicts their success. *Proceedings of the National Academy of Sciences*, 118(26), e2011695118.

### Annotations

#### 1. Purposes

##### a. Motivation (one-sentence summary)

To quantify the semantic progression of texts ((i.e., speed, volume, and circuitousness) and examine how such measures relate to success in different domains.

##### b. Specific Rationale (several bullet points)

Attempts to answer why some narratives, or other types of discourse are more successful, were hindered by quantification.

Narratives are often described as speed, the amount it covers, or going in circles, but little work has quantified such features or examined whether they might explain the success of a document.

It is because measuring features of text contents is difficult. Human coding cannot address big data. Also, it is often subjective and difficult to scale, even if it is possible for a small number of texts.

This study addresses this gap using natural language processing and machine learning plot chunks of texts as sequential points in a multidimensional space and quantify features of the semantic path (i.e., speed, volume, and circuitousness)

##### c. Research Questions and/or Hypotheses

Why are some narratives, or other types of discourse, more successful?

And could a simple set of measures help explain variation in success in different domains?

#### 2. Method

##### a. Research Design

- Computational text analysis using NLP

##### b. Sample

Text data of movies, television shows, and academic papers, these domains were chosen based on data availability

Over 4,000 movies, 12,000 TV show episodes, citations of 29,000 academic papers published in 22 journals

### c. Measures

**Simple set of measures: semantic path (i.e., speed, volume, and circuitousness).**

**-Speed:** content can move slower or faster. They measure the distance texts travel between consecutive chunks for speed. Word embeddings capture semantic similarity so consecutive chunks that are farther away are more likely to discuss different topics or themes.

**-Volume:** How much content covers ground or how many themes it touches. They measure such volume by approximating points with an ellipsoid by solving an optimization problem that finds the minimum volume ellipsoid containing all of these points

**-Circuitousness:** it is defined as the ratio of actual distance traveled to the shortest possible path.

### Success.

**-Success of movies:** movie Database (IMDB) ratings of 4,118 movies based on their subtitles.

**-Success of TV shows:** IMDB ratings of 12,401 episodes of TV shows based on closed captions.

**-Success of academic papers:** citations received by 29,300 academic articles published in 22 different journals in psychology, economics, sociology, political science, and anthropology between 1990 and 2019.

### d. Analysis

- Take texts (e.g., movies or academic papers), break them into approximately equal-sized chunks or windows (about 250 words),

- Using word embeddings, which is a method that turns words into high-dimensional numerical vectors, so that the distances between the numbers reflect the meanings of the words, they plot each chunk in a high-dimensional semantic space, and examine the path between chunks.

-They calculate new measures that characterize each text's semantic path (speed, volume, and Circuitousness) .

-They examine the relationship between these measures and success in three domains (i.e., movies, television shows, and academic papers) using Lasso regression

-They control for textual content by including 100 topic intensities estimated by latent Dirichlet allocation to strengthen their analysis ensuring that success is not influenced by certain topics.

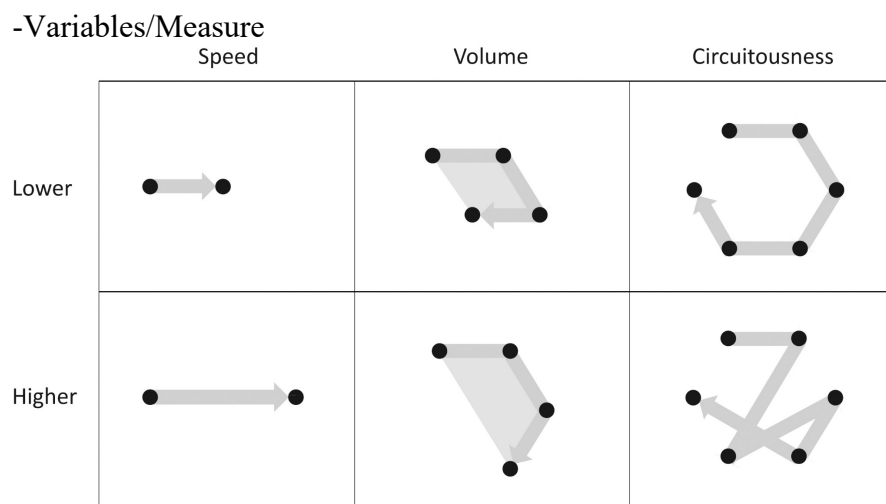
## 3. Results and Conclusions

### a. Answers to RQs and Hypotheses

-Narratives that move faster (i.e., travel farther in consecutive periods, on average) are evaluated more favorably for movies and TV shows

- TV show episodes that cover less volume are also evaluated more favorably.
- Academic [apers that move faster are cited less.
- Volume has the opposite effect in papers compared to TV shows, papers that cover more ground are cited more
- Papers that are more circuitous receive more citations.

## b. Include Key Figures/Tables



## -Results

**Table 1. Link between semantic progression and success**

|                            | Movies | TV show episodes | Academic papers |
|----------------------------|--------|------------------|-----------------|
| Average speed              | 0.048* | 0.072*           | -0.125*         |
| Normalized volume          | 0      | -0.082*          | 0.095*          |
| Circuitousness             | 0      | 0.006            | 0.070*          |
| Controls                   |        |                  |                 |
| Year fixed effects         | Yes    |                  | Yes             |
| Genre fixed effects        | Yes    | Yes              |                 |
| Movie duration             | Yes    |                  |                 |
| TV channels fixed effects  |        | Yes              |                 |
| Journal fixed effects      |        |                  | Yes             |
| No. of pages               |        |                  | Yes             |
| Log(words in document)     | Yes    | Yes              | Yes             |
| Log(sentences in document) | Yes    | Yes              | Yes             |
| Topic intensities          | Yes    | Yes              | Yes             |
| No. of parameters          | 169    | 148              | 158             |
| No. of observations        | 4,118  | 12,336           | 29,300          |
| Mean-squared error         | 0.711  | 0.793            | 1.066           |
| R <sup>2</sup>             | 0.306  | 0.326            | 0.364           |

Note that all independent variables for which coefficients are reported are standardized. The dependent variable is not standardized. Parameters are estimated using a lasso regression. Confidence intervals are obtained via bootstrapping. \*The 95% confidence interval does not include 0. Dependent variable is IMDB ratings for movies and TV show episodes and log(1 + citations) for academic papers.

### **c. Authors' Conclusions**

- Narratives and other forms of discourse offer valuable opportunities to examine the content features that influence success, as well as their underlying psychological mechanisms.
- Advances in natural language processing can offer wide range of new directions for future research.

## **4. Discussion**

### **a. Implications**

- It contributes to cultural analytics and understanding why cultural items succeed and fail, suggesting success is not completely random and that some features may also play an important role.
- It also highlights the value of natural language processing to study culture, as NLP can address a key challenge on quantifying narratives and cultural dynamics.
- It opens up opportunities for further research in many areas.
- It supports recent work on how individuals' psychological processes shape collective outcome because how people process and evaluate content, may shape success of content.

### **b. Limitations**

They did not explicitly mention limitations.

Future Directions

- Future study can explore cross-domain differences, the type of discourse (e.g., narrative vs. exposition), goal (e.g., to entertain vs. impart knowledge), modality (e.g., video vs. written), outcome measure (e.g., liking vs. citations), audience expectations, and other types of texts (e.g., books, speeches, or documentaries).
- Future work could explore the cognitive and social processes that explain the findings of this study.
- Work might also examine the outcomes of these features for other downstream outcomes (e.g., comprehension, memory, and persuasion).

--- Page Break Here ---  
Response Letter (~600 words)

- Your assessment of the study

As the authors argue, I believe this study would benefit many future studies. I believe that proposing a way to quantify content that includes narratives is a groundbreaking and meaningful contribution. Some may consider the fact that the study does not consider qualitative features, sentiment, or topics as a limitation. However, I think the key contribution is that the study presents a universally adaptable method that can extract features only through quantitative text analysis, even without considering additional factors.

I also found it very smart that the study proposes three features, which are speed, volume, and circuitousness, to capture the features of documents or narratives that cannot be measured by a single variable. I think this approach using three variables makes the analysis to more inclusively represent the complexity of narratives.

Furthermore, after reading the appendix, I saw that the study includes many additional analyses to show that the findings are consistent across different specifications, which increases the validity of the results and strengthens the overall findings. I also think the implications of the study are further enhanced by the fact that it analyzes not only academic papers, but also movies and TV shows.

- Weaknesses

However, I still have several questions. The paper mentions that “*narratives usually involve temporality, or a sequence of events and actions, and that similar questions could also be asked of other types of discourse*”. It also notes that narratives in TV shows and movies are different from those in academic papers. Given this, I am not fully convinced that the quantified semantic progression constructed in this study is an appropriate operationalization of the features of narratives across these different domains.

Although proposing this method itself is a very meaningful contribution and they show that their quantified features predict success across TV, movies, and paper, I am skeptical about whether these methods can be used to predict the success of individual pieces of content. While “success” is a main research topic of the paper, I am curious whether these quantified features might be better suited to addressing other research questions other than predicting success.

I am also curious about whether controlling the length sufficiently addresses its influence. Even if the average volume of different documents is similar, people find long documents overwhelming, no matter how much the average is.